

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is:

The web server was experiencing a TCP SYN Flood attack that caused it to become overloaded and unable to respond to legitimate user requests.

The logs show that:

The network logs show a large number of TCP SYN packets coming from the IP address 203.0.113.0. The server initially responded to these requests, but the attacker continued sending SYN packets at a very high rate. As the attack continued, legitimate users started receiving timeout and connection reset errors. Eventually, the server stopped responding to normal traffic.

This event could be:

This event is a Denial of Service (DoS) attack, specifically a SYN Flood attack. Since the attack originated from a single IP address, it is classified as a DoS attack rather than a Distributed Denial of Service (DDoS) attack.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

1. The client sends a SYN packet to the server requesting a connection.
2. The server responds with a SYN-ACK packet to acknowledge the request.

3.The client sends an ACK packet back to the server, completing the connection.

Explain what happens when a malicious actor sends a large number of SYN packets all at once:

When a malicious actor sends thousands of SYN packets, the server attempts to respond to each request and reserves resources for incomplete connections. As the number of requests increases, the server's resources become exhausted, preventing it from handling legitimate traffic.

Explain what the logs indicate and how that affects the server:

The logs indicate that the attacker continuously sent SYN packets to the web server without establishing normal connections. This caused the server to become overwhelmed and unable to process legitimate user requests. As a result, employees and customers experienced connection timeouts, failed webpage requests, and website unavailability. This attack negatively impacted business operations because users could no longer access the company's sales webpage.

Potential Recommendations:

To prevent similar attacks in the future, the organization should implement SYN flood protection, configure rate limiting on the firewall, deploy intrusion prevention systems (IPS), and continuously monitor network traffic for suspicious activity.